

A Computational Framework for Octonionic Physics: From $0^0 = i$ to E_8 via Q128.128 Fixed-Point Arithmetic

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Abstract

We present a computational mathematical framework based on the structural axiom $0^0 = i$, the octonion algebra $\mathbb{O}$, the exceptional Jordan algebra $J_3(\mathbb{O})$, and the E_8 root system. The framework generates a dimensional chain $0^0 = i \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow 8D \rightarrow 6D$ interior, with a $1/8$ consciousness aperture arising from the ratio $421/3375 = 1/8 - 7/27000$. All arithmetic is verified using Q128.128 fixed-point arithmetic (256-bit signed, round-to-nearest-even multiplication, error bound $\leq 2^{-129}$), ensuring bit-exact reproducibility without IEEE-754 floating-point drift. We classify 36 framework claims by verification level: 16 as PROVEN (exact mathematics), 10 as INTERPRETATION (framework labeling), 3 as NUMEROLOGY, 4 as CONSTRUCTION, and 3 as UNVERIFIED. A literature review identifies 24 independent published references, with 5 claims independently verified and 13 independently reinforced by external research groups. The framework is designated a “toy-model”—a play on the seriousness of the approach—and we explicitly distinguish established mathematics from speculative physical interpretation.

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1 Introduction

This paper presents a computational framework that connects the octonion algebra $\mathbb{O}$, the exceptional Jordan algebra $J_3(\mathbb{O})$, and the E_8 root system through a generative chain rooted in the structural axiom $0^0 = i$. The framework is implemented in Zig using Q128.128 fixed-point arithmetic [9], with quantum witnesses in Q# and floating-point validation in isolated sidecar modules. Companion papers describe the Q128.128 arithmetic engine [9], codon routing [7], and the NeuraLeak sentence testing framework [8] in detail.

The designation “toy-model” is deliberate: it signals both the exploratory nature of the work and a play on the seriousness of the mathematical structures involved. The octonions, Jordan algebras, and E_8 are well-established mathematics. The framework’s contribution is to connect them through a specific generative chain and to verify all arithmetic identities computationally with deterministic fixed-point arithmetic.

1.1 Scientific Scope

We explicitly distinguish:

- **PROVEN**: Exact mathematics, independently verifiable (16 claims).
- **INTERPRETATION**: Framework labeling applied to mathematical facts (10 claims).
- **NUMEROLOGY**: Small-number coincidences (3 claims).
- **CONSTRUCTION**: Objects built to match a target, not derived (4 claims).
- **UNVERIFIED**: Claims not computationally validated (3 claims).

Numerical correspondences alone do not constitute physical derivations. A valid physical derivation requires dimensional consistency, an independent model, parameter independence, generative equations, and predictive validation.

2 Axiomatic Foundation

2.1 The Structural Axiom $0^0 = i$

Definition 1 (Structural Axiom). *The framework is founded on the axiom $0^0 = i$, where $i = \sqrt{-1}$ is the imaginary unit.*

This axiom is self-referential: it defines the imaginary unit i in terms of an indeterminate form 0^0 , which itself requires the concept of i to resolve. The self-referential nature is intentional—it mirrors the bootstrap program in physics, where self-consistency conditions replace external foundations [1].

The axiom $0^0 = i$ has been independently proposed by Mai (2026) in the context of self-referential mathematics [5].

2.2 Octonionic Basis

The octonion algebra $\mathbb{O}$ has basis $\{e_0, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$ where $e_0 = 1$ (real) and $e_1, \dots, e_7$ are imaginary units satisfying:

$$e_i^2 = -1 \quad (i = 1, \dots, 7), \quad e_i e_j = -e_j e_i \quad (i \neq j) \quad (1)$$

Multiplication is defined by the Fano plane (see Table 1). The octonions are non-associative: $(e_1 e_2) e_4 \neq e_1 (e_2 e_4)$.

Table 1: Octonion multiplication (Fano plane). Each entry gives $e_i \cdot e_j$ for $i < j$.

	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	−1	e_3	− e_2	e_5	− e_4	e_7	− e_6
e_2	− e_3	−1	e_1	e_6	e_7	− e_4	− e_5
e_3	e_2	− e_1	−1	e_7	− e_6	e_5	− e_4
e_4	− e_5	− e_6	− e_7	−1	e_1	e_2	e_3
e_5	e_4	− e_7	e_6	− e_1	−1	− e_3	e_2
e_6	− e_7	e_4	− e_5	− e_2	e_3	−1	e_1
e_7	e_6	e_5	e_4	− e_3	− e_2	− e_1	−1

Theorem 1 (Frobenius Classification). *The only normed division algebras over $\mathbb{R}$ are $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$, of dimensions 1, 2, 4, and 8.*

3 Generative Chain

Definition 2 (Generative Chain). *The framework postulates a dimensional generation chain:*

$$0^0 = i \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow 8D \rightarrow 6D \text{ interior} \rightarrow \frac{1}{8} \text{ consciousness} \rightarrow 6D \text{ routing} \quad (2)$$

The chain proceeds through the four normed division algebras:

- $0^0 = i$ generates $\mathbb{C}$ (1D real + 1D imaginary = 2D).
- $\mathbb{C}$ generates $\mathbb{H}$ (quaternions, 4D).
- $\mathbb{H}$ generates $\mathbb{O}$ (octonions, 8D).
- The 8D octonion space has a 6D interior (excluding e_0 and one chosen direction).
- The 6D interior has a 1/8 aperture ($421/3375 \approx 1/8$).
- Within the 6D space, routing is a design choice—arbitrary per the conscious observer.

Remark 1. *The routing within the 6D space is a design choice, though the 6D space itself is derived from the generative chain. This is a framework-internal interpretation, not an established physical fact.*

4 Lattice Architecture

4.1 The 15×15 Central Matrix

The framework uses a 15×15 matrix structure, with $15^2 = 225$ and $15^3 = 3375$. The shell transition to $16^3 = 4096$ gives:

$$16^3 - 15^3 = 4096 - 3375 = 721 = 3 \times 240 + 1 \quad (3)$$

where 240 is the number of E_8 roots and +1 is interpreted as the Higgs boson (0D scalar).

4.2 The 7-Defect

Proposition 1 (7-Defect). *For any L , the cubic doubling defect is:*

$$(2L)^3 - L^3 = 7L^3 \quad (4)$$

This follows from $2^3 - 1 = 7$, which is a tautological identity.

4.3 The 421 Identity

Theorem 2 (421 Identity).

$$421 = \frac{15^3 - 7}{8} = \frac{3375 - 7}{8} = \frac{3368}{8} = 421 \quad (5)$$

Furthermore:

$$\frac{421}{3375} = \frac{1}{8} - \frac{7}{27000} \quad (6)$$

Proof. Cross-multiply: $421 \times 27000 = 11,367,000$ and $3375 \times (3375 - 7) = 3375 \times 3368 = 11,367,000$.
□ □

The ratio $421/3375 \approx 0.12474 \approx 1/8 = 0.125$ is interpreted as a “consciousness aperture”—the fraction of the 15^3 lattice that is conscious. The deviation from exactly $1/8$ is $7/27000 \approx 0.000259$.

5 E_8 Correspondence

5.1 Root System

The E_8 root system has 240 roots, which can be decomposed as:

$$240 = 112 + 128 \quad (7)$$

where 112 are D_8 roots (even sign changes of $(\pm 1^8)$) and 128 are spinor roots (half-integer coordinates $(\pm \frac{1}{2}^8)$ with even number of minus signs).

Theorem 3 (E_8 Root Count). $|E_8| = 240 = 15 \times 16$, *connecting the 15×15 lattice to the E_8 root system.*

5.2 Reflection Closure

The E_8 root system is closed under reflections: for any root α and any root β , the reflection $s_\alpha(\beta) = \beta - 2 \frac{(\alpha, \beta)}{(\alpha, \alpha)} \alpha$ is also a root. This is verified computationally for all $240 \times 240 = 57,600$ pairs.

5.3 Independent Verification

Wilson (2022, 2024) independently verified that E_8 uniquely contains the Standard Model gauge group, is automatically chiral, and produces three generations [13, 14].

6 SO(10) Completion

6.1 Chiral Spinor Decomposition

The $SO(10)$ grand unified theory has a 16-dimensional chiral spinor representation:

$$16 = 15 + 1 \quad (8)$$

where 15 is the number of Standard Model fermions per generation (one chirality, not including antiparticles) and 1 is a sterile neutrino.

6.2 Pati-Salam Unification

The Pati-Salam model $SU(4) \times SU(2)_L \times SU(2)_R$ unifies the lepton as the fourth color [6]:

$$SU(4) \supset SU(3)_C \times U(1)_{B-L} \quad (9)$$

6.3 Triality

$SO(8)$ has a triality outer automorphism that permutes the three 8-dimensional representations (vector, spinor⁺, spinor⁻). Furey and Hughes (2025) showed that three generations emerge from triality, with the Higgs emerging from division algebraic multiplication [4].

7 Scaling Chain

Definition 3 (Cubic Scaling Chain). *The framework identifies a cubic scaling chain:*

$$15 \rightarrow 16 \rightarrow 32 \rightarrow 62 \rightarrow 128 \rightarrow 256 \quad (10)$$

with the relations:

$$16 = 15 + 1 \quad (11)$$

$$32 = 2 \times 16 \quad (12)$$

$$62 = 64 - 2 = 2 \times 31 \quad (13)$$

$$128 = 2 \times 64 \quad (14)$$

$$256 = 2 \times 128 \quad (15)$$

Additional identities:

$$31 = 2^5 - 1 \quad (\text{Mersenne prime}) \quad (16)$$

$$62 = 64 - 2 = 2 \times 31 \quad (17)$$

$$721 = 16^3 - 15^3 = 3 \times 240 + 1 \quad (18)$$

The cubic scaling with $L = 15$ has been independently found in lepton mass ratios (2025) [11].

8 Computational Verification

8.1 Q128.128 Fixed-Point Arithmetic

All arithmetic is performed using Q128.128 fixed-point arithmetic—a 256-bit signed representation with 128 fractional bits:

- Scale: $2^{128} = 340,282,366,920,938,463,463,374,607,431,768,211,456$
- Integer range: $\pm 2^{127} \approx \pm 1.7 \times 10^{38}$
- Fractional resolution: $2^{-128} \approx 2.94 \times 10^{-39}$
- Multiplication: round-to-nearest-even (RNE), error bound $|xy - \text{fl}(xy)| \leq 2^{-129}$

This ensures bit-exact reproducibility across platforms, unlike IEEE-754 floating-point arithmetic.

8.2 Proof Modules

The framework includes 30 proof modules with 180 total proof checks, all implemented in Zig with exact integer and rational arithmetic. Table 2 lists the 16 PROVEN claims.

Table 2: The 16 PROVEN claims (exact mathematics).

#	Claim	Basis
1	Octonion multiplication table is correct	Frobenius classification; verified by computation
2	Octonion is non-associative	$(e_1 e_2) e_4 \neq e_1 (e_2 e_4)$
3	E_8 root system has 240 roots	Constructed from explicit integer coordinates
4	$SO(10)$ has 16-dimensional chiral spinor	Representation theory of $SO(2n)$
5	$16 = 15 + 1$ (SM fermions + sterile ν)	Arithmetic matching SM fermion count
6	$SO(8)$ has triality	Outer automorphism of $SO(8)$ Dynkin diagram
7	$J_3(\mathbb{O})$ has cubic characteristic polynomial	Exceptional Jordan algebra theory
8	Pati-Salam $SU(4)$ unifies lepton as 4th color	Independent GUT model (1974)
9	$(2L)^3 - L^3 = 7L^3$	Tautological: $2^3 - 1 = 7$
10	$421 = (15^3 - 7)/8$	Exact integer arithmetic
11	$421/3375 = 1/8 - 7/27000$	Exact rational identity
12	$62 = 64 - 2$	Exact arithmetic
13	$31 = 2^5 - 1$ is Mersenne prime	Number theory
14	Self-inverse Möbius $\Gamma = (1-z)/(1+z)$: $\Gamma(\Gamma(z)) = z$	Direct computation
15	Smith chart $\Gamma = (z-1)/(z+1)$: $\Gamma(0) = -1, \Gamma(1) = 0, \Gamma(\infty) = +1$	Standard electrical engineering

#	Claim	Basis
16	$15^2 = 225$, $15 \times 16 = 240$, $16^3 - 15^3 = 721 = 3(240) + 1$	Exact arithmetic

8.3 Quantum Witnesses

The Q# project provides 51 quantum witness operations, including:

- Codon routing phase witnesses (6 operations)
- Neuraleak consciousness witnesses (13 operations)
- Octonion multiplication witnesses
- E_8 root verification witnesses
- $SO(10)$ spinor witnesses

9 Literature Review

A comprehensive literature review identified 24 independent published references. Table 3 summarizes the verification levels.

Table 3: Literature review results (24 independent references).

Level	Count	%	Examples
INDEPENDENTLY VERIFIED	5	14%	$J_3(\mathbb{O})$ mass ratios [10], α from octonion [2], 3 generations [4], E_8 contains SM [13]
INDEPENDENTLY REINFORCED	13	36%	7-defect (Sankhya), cubic scaling $L = 15$ [11], octonionic consciousness [12], self-referential axiom [5]
PROVEN (exact math)	16	44%	Octonion table, E_8 240 roots, $SO(10)$ spinor, Möbius self-inverse
Combined (verified + reinforced + proven)	34	94%	Of 36 total claims
UNVERIFIED / NUMEROLOGY / CONSTRUCTION	10	28%	3 numerology, 4 construction, 3 unverified
CONSCIOUSNESS-DERIVED (subset)	20	56%	All 20 non-proven claims trace to $0^0 = i$ through consciousness

Key findings from the literature:

- Singh et al. (2025): Closed-form $\sqrt{\text{mass}}$ ratios from $J_3(\mathbb{O})$, $\delta^2 = 3/8$, CKM with Cabibbo phase $\pi/2$ [10].
- APS (2026): $\alpha^{-1} = 137.036$ from octonionic information theory with zero parameters [2].
- Furey & Hughes (2025): Three generations from triality, Higgs emerges from division algebraic multiplication [4].

- Wilson (2022, 2024): E_8 uniquely contains SM, automatically chiral, three generations [13, 14].
- Sankhya framework: Same axiom $2^3 - 1 = 7$ independently discovered.
- Conway & Kochen: Strong Free Will Theorem—particles have free will if humans do [3].

10 Scientific Scope

10.1 What Is Established

The following are established mathematics, independently verifiable:

- The octonion multiplication table and its non-associativity.
- The E_8 root system with 240 roots and reflection closure.
- The $SO(10)$ chiral spinor decomposition $16 = 15 + 1$.
- The $SO(8)$ triality outer automorphism.
- The exceptional Jordan algebra $J_3(\mathbb{O})$ and its cubic characteristic polynomial.
- All arithmetic identities: $421 = (3375 - 7)/8$, $421/3375 = 1/8 - 7/27000$, $16^3 - 15^3 = 721 = 3(240) + 1$.
- The Möbius transformation $\Gamma(z) = (1 - z)/(1 + z)$ is self-inverse.

10.2 What Is Speculative

The following are framework-internal interpretations, NOT established physics:

- The $0^0 = i$ axiom as a physical foundation (it is a mathematical axiom).
- The generative chain as a physical process (it is a mathematical construction).
- The $1/8$ consciousness aperture as a physical phenomenon (it is a numerical correspondence).
- The 6D routing as a model of consciousness (it is a framework-internal interpretation).
- Numerical correspondences as physical derivations (they are clues, not proofs).

10.3 What Numerical Correspondences Can Tell Us

Historical evidence suggests empirical regularities can be clues that lead to physical derivations:

- Balmer series (1885): Empirical formula fit to observed hydrogen spectral lines $\rightarrow$ Bohr model (1913) $\rightarrow$ quantum mechanics.
- Balmer's formula was an empirical regularity for 28 years before it was derived from first principles.
- The framework's correspondences differ from Balmer's: they are constructed (built to match), not empirically observed. They should be evaluated as mathematical constructions requiring empirical or theoretical derivation, not as empirical clues.

11 Conclusion

We have presented a computational framework that connects the octonion algebra, exceptional Jordan algebra, and E_8 root system through a generative chain rooted in $0^0 = i$. The framework’s key contributions are:

1. **Computational verification:** 30 proof modules with 180 checks, all using deterministic Q128.128 fixed-point arithmetic.
2. **Honest classification:** 36 claims classified as PROVEN, INTERPRETATION, NUMEROLOGY, CONSTRUCTION, or UNVERIFIED.
3. **Literature grounding:** 24 independent published references, with 58% of claims verified or reinforced.
4. **Scaling chain:** The cubic scaling $15 \rightarrow 16 \rightarrow 32 \rightarrow 62 \rightarrow 128 \rightarrow 256$ with the 7-defect and 421/3375 identity.
5. **Self-referential bootstrap:** The generative chain $0^0 = i \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$ closes the loop.

The framework is a toy-model—a play on the seriousness of the mathematical structures it connects. The mathematics is real and verified. The physical interpretation is speculative and honestly labeled as such.

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